

Daniel Ge – BCW Global Solutions

Problem 1 (BAMO 2013/4)

Claim: The only n that works are the perfect squares greater than 2.

Proof:

We first prove by construction that all perfect squares greater than 2 work. For a given $n = k^2$ where k is an integer greater than 1, we flip only the first $k - 1$ fractions s.t. the final the product would be,

$$\left(\frac{1}{2} \cdot \frac{2}{3} \cdot \dots \cdot \frac{k-1}{k}\right) \cdot \left(\frac{k+1}{k} \cdot \frac{k+2}{k+1} \cdot \dots \cdot \frac{n}{n-1}\right)$$

Doing the cancelation, we get,

$$= \frac{1}{k} \cdot \frac{n}{k} = \frac{n}{k^2} = 1$$

We now prove that there are no more n that work. For the sake of contradiction, let this be possible for a non-perfect square $n > 2$. We can write the flipped product as,

$$\prod \frac{a_i}{b_i} = 1$$

Therefore, $\prod a_i b_i$ is a perfect square. However, $\prod a_i b_i = 1 \cdot 2^2 \cdot 3^2 \cdot \dots \cdot n$. This value cannot be a perfect square, so we have a contradiction and are done.

Problem 2 (HMMT 2013)

We write $c_1, c_2, \dots, c_{2013}$ as the number of occurrences of $1, 2, \dots, 2023$ in a respectively. We complementary count for the expected number of 0 in $c_1, c_2, \dots, c_{2013}$.

The probability that $c_i = 0$ is the same over all i and is simply $\left(\frac{2012}{2013}\right)^{2013}$. By linearity of expectations, the expected number of 0s is $\frac{2012^{2013}}{2013^{2012}}$. The grand total is hence $2013 - \frac{2012^{2013}}{2013^{2012}}$. $\square$

Problem 3 (IMC 2002)

For a random pair of contestants, we find the expected value of the problems solved by either of them using complementary counting.

$$E = 6 \left(1 - \left(\frac{2}{5}\right)^2\right) = 5 + \frac{1}{25}$$

Therefore, by Pigeonhole, there is a pair where this value is 6. $\square$

Problem 4 (Tournament of Towns 2001)

Sketch:

For contradiction, let the rooks still be non-attacking at the end.

Let $\{x_1, x_2, \dots, x_{15}\}$ and $\{y_1, y_2, \dots, y_{15}\}$ be the coordinates of the rooks. Whenever the rooks are non-attacking, these are $\{1, \dots, 15\}$. Therefore, at the start and end,

$$\sum x_i = \sum y_i = 1 + 2 + \dots + 15 \Rightarrow \sum x_i + \sum y_i \equiv 0 \pmod{2}$$

During each of the 15 moves, the parity of $\sum x_i + \sum y_i$ must change. Therefore, at the end,

$$\sum x_i + \sum y_i \equiv 1 \pmod{2}$$

Therefore, we have contradiction. $\square$

Problem 5 (PAGMO 2022/1)

We can instead count the sum that each pair of *friends* generates. If the *friends* are the same color, that pair adds 0 to the sum. If not, that pair adds 3 to the sum.

Clearly, we want to minimize the number of identical pairs of *friends*. One such coloring is as follows:

R	B	G	R	B	G	R	B	G
B	G	R	B	G	R	B	G	R
G	R	B	G	R	B	G	R	B
R	B	G	R	B	G	R	B	G
B	G	R	B	G	R	B	G	R
G	R	B	G	R	B	G	R	B
R	B	G	R	B	G	R	B	G
B	G	R	B	G	R	B	G	R
G	R	B	G	R	B	G	R	B

In this case, all pairs of friends are distinct, so we have the maximum. There are $2 \cdot 9 \cdot 9 = 162$ pairs of friends so the total score is $162 \cdot 3 = 486$. $\square$

12 clubs, 5.5 hours cumulative (including examples)

Problem 6 (AIME 1985)

We can count the total amount of points earned. Each match earns a total of one point.

We call the bottom 10 players the *n00bs* and the other players the *pr0s*.

We let x be the number of *pr0s*.

The *n00bs* scored a total of $\frac{10 \cdot 9}{2} = 45$ points against *n00bs*. Since each of the *n00bs* an equal amount against *n00bs* as *pr0s*, a total of 45 points were scored by *n00bs* against *pr0s* out of $10x$ *n00b-pr0* pairings.

The other $10x - 45$ points were scored by the *pr0s* against the *n00bs*. Since the *pr0s* scored an equal amount against *n00bs* as other *pr0s*, $10x - 45$ points were scored in *pr0-pr0* pairings. Therefore,

$$\frac{x(x-1)}{2} = 10x - 45 \Leftrightarrow x^2 - x = 20x - 90 \Leftrightarrow x^2 - 21x + 90 = 0 \Leftrightarrow x = 6, 15$$

If $x = 6$, then 15 points were scored by the *pr0s* against the *n00bs* (2.5 average) while 45 the other way (4.5 average) around, which would mean the *pr0s* are worse than the *n00bs*, which doesn't make sense.

Therefore, the only solution is $x = 15$ for a total of 25 players in the tournament. $\square$

Problem 7 (Russia 1996)

We find the expected number of overlap when picking two committees at random. We let x_i be the number of committees the i -th person is on. Note that,

$$\sum x_i = 16000 \cdot 80$$

The total number of intersections is $\sum \binom{x_i}{2}$. By Jensen's, this is minimized when $x_i = 800$ for all x , meaning there are at least a total of $\frac{800 \cdot 799 \cdot 1600}{2}$ intersections over all $\frac{16000 \cdot 15999}{2}$ pairs of committees. This means that the expected value is $\frac{800 \cdot 799 \cdot 1600}{16000 \cdot 15999} > 3$ intersections. By Pigeonhole, we are done. $\square$

18 clubs, 6.5 hours cumulative

Problem 9 (Iberoamerican 2001/3)

If we can prove this with each X having exactly r elements, then it is true for some X having more than r elements, so we'll assume each X to have exactly r elements in this proof.

We double count the number of triples (x, A, B) where A and B are distinct subsets of X , and x is in both A and B . We denote this count as N .

For the sake of contradiction, if no two subsets have at least $r - \frac{nk}{4(k-1)}$ elements intersected, then,

$$N < \binom{k}{2} \left(r - \frac{nk}{4(k-1)} \right)$$

We let x_i be the number of subsets the i -th element is in.

$$\sum x_i = rk$$

The total number of intersections is $\sum \binom{x_i}{2}$. By Jensen's, minimized when $x_i = \frac{rk}{n}$.

$$\begin{aligned}
N &\geq n \cdot \binom{\frac{rk}{n}}{2} \\
&\Leftrightarrow n \cdot \binom{\frac{rk}{n}}{2} < \binom{k}{2} \left(r - \frac{nk}{4(k-1)} \right) \\
&\Leftrightarrow n \cdot \frac{rk}{n} \cdot \left(\frac{rk}{n} - 1 \right) < k(k-1) \left(r - \frac{nk}{4(k-1)} \right) \\
&\Leftrightarrow \frac{r^2 k^2}{n} - rk < rk(k-1) - \frac{nk^2}{4} \\
&\Leftrightarrow \frac{r^2 k}{n} - r < rk - r - \frac{nk}{4} \\
&\Leftrightarrow \frac{r^2}{n} < r - \frac{n}{4} \\
&\Leftrightarrow r^2 - nr + \frac{n^2}{4} < 0 \\
&\Leftrightarrow \left(r - \frac{n}{2} \right)^2 < 0 \quad (!)
\end{aligned}$$

Problem 10 (Korea 2005)

Let n be the number of questions on the test and x be the total number triples (s, i, j) where student s solved problem i but not problem j and $i \neq j$.

We are given by the problem statement that for any pair (i, j) , we have at least 6 of s .

$$x \geq 6 \binom{n}{2}$$

We now count by each student. Let m be the number of solves of a student s_i . The number of (s_i, i, j) is hence $m(n-m)$. By AM-GM, $m(n-m) \leq \frac{n^2}{4}$. Since there are 11 students,

$$x \leq \frac{11n^2}{4}$$

Therefore,

$$6 \binom{n}{2} \leq \frac{11n^2}{4} \Leftrightarrow 12n(n-1) \leq 11n^2 \Leftrightarrow n^2 \leq 12n$$

Since $n \geq 0$,

$$\Leftrightarrow n \leq 12 \quad \square$$

28 clubs, 7 hours cumulative

Problem 11 (USAMO 1999/1)

Possible Construction: Fill the entire board with checkers.

[SKETCH] *Proof of at least $\frac{n^2-2}{3}$:*

Let a be the number of squares that either have checkers or are adjacent to checkers. We prove $a \leq 3(\# \text{ of checkers}) + 2$ and we're done.

Since all the checkers are connected, we re-construct this patch of x checkers one at a time. The first checker grants us at most $a = 5$. Each checker afterwards can only increment a by at most 3 so done.

Problem 13 (Hong Kong TST 2019/1/5)

We claim that this is impossible.

For the sake of contradiction, let there be 2019 such planes. We let $x_1, x_2, \dots, x_{2019}$ be the number of points that are on each plane.

We double count the number of (not-necessarily distinct) planes formed with each triplet of points.

There are $\binom{24}{3} = 2024$ ways to select 3 points from the 24, so that's the amount we should have.

Therefore,

$$\sum \binom{x_i}{3} = 2024$$

If $x_i > 4$ for any i , then we would have at least 10 triplets of points that all correspond to the same plane, meaning we can have at most 2015 distinct planes. Therefore, $x_i = 3$ or $x_i = 4$.

We let a be the number of x_i where $x_i = 3$, and b be the number of x_i where $x_i = 4$. We get the system of equations,

$$a + 4b = 2024; a + b = 2019$$

There are no integer solutions meaning there's a contradiction. We're done.

Problem 14 (IMO 2001/4)

For the sake of contradiction, let there exist no permutations $a \neq b$ such that $S(a) - S(b)$ is a divisor of $n!$. That means there are no two permutations a, b where $S(a) = S(b)$. This means $S(a)$ over all permutations a must make up the set $\{0, 1, \dots, n! - 1\}$.

We double count the sum of $x := S(a)$. We know $S(a) \bmod n!$ is $\frac{(n!-1)n!}{2} = \frac{n!}{2}(n! - 1) \equiv \frac{n!}{2} \bmod n!$ since $n! - 1$ is odd.

We can also count how much each c_i contributes to x :

$$x = \left(\sum c_i \right) (n-1)! (1+2+\dots+n) = \frac{(\sum c_i)(n+1)!}{2}$$

Since $n+1$ is even, $\frac{(\sum c_i)(n+1)!}{2} \equiv 0 \pmod{n!}$, which is a contradiction.

Problem 15 (BAMO 2017/4)

For these rectangles to cover the interior of P , h must be at least the maximal distance to the closest side over all the points inside P . We let h must be this maximal distance to the closest side over all the points inside P .

Let $d_1, d_2, \dots, d_n$ be the distance of this point to each of the sides of the n -gon. We let $x_1, \dots, x_n$ be the length of these sides. Obviously, the area of P ,

$$[P] = \frac{d_1 x_1 + d_2 x_2 + \dots + d_n x_n}{2}$$

The total area A of the rectangles is,

$$A = d_1 h + d_2 h + \dots + d_n h$$

Obviously, $2[P] \geq A$ since $\min(d_1, \dots, d_n) = h \Rightarrow d_1, d_2, \dots, d_n \leq h$.

Problem 17 (Romania 2004)

We let $z_i = a_i + i b_i$.

$$\sum |z_i|^2 = 1 \Leftrightarrow \sum a_i^2 + \sum b_i^2 = 1$$

We want that,

$$\begin{aligned} \left| \sum \varepsilon_i z_i \right| \leq 1 &\Leftrightarrow \left| \sum \varepsilon_i z_i \right|^2 \leq 1 \\ &\Leftrightarrow \left(\sum \varepsilon_i a_i \right)^2 + \left(\sum \varepsilon_i b_i \right)^2 \leq \sum a_i^2 + \sum b_i^2 \\ &\Leftrightarrow \sum \varepsilon_i^2 a_i^2 + \sum \varepsilon_i^2 b_i^2 + \sum_{sym} \varepsilon_1 \varepsilon_2 a_1 a_2 + \sum_{sym} \varepsilon_1 \varepsilon_2 b_1 b_2 \leq \sum a_i^2 + \sum b_i^2 \\ &\Leftrightarrow \sum_{sym} \varepsilon_1 \varepsilon_2 a_1 a_2 + \sum_{sym} \varepsilon_1 \varepsilon_2 b_1 b_2 \leq 0 \end{aligned}$$

Claim: The expected value of $\sum_{sym} \varepsilon_1 \varepsilon_2 a_1 a_2 + \sum_{sym} \varepsilon_1 \varepsilon_2 b_1 b_2$ over all selections of ε is zero.

Proof: For each pair $a_i a_j$, $\varepsilon_i \varepsilon_j$ is positive 1 the same number of times as it is negative 1.

Therefore, by Pigeonhole, there must be such a valid configuration of ε .

45 clubs, 10.5 hours cumulative